

Assignment # 1

Retail Sales Dashboard Assignment (Excel Pivot Tables & Pivot Charts)

You are working as a BI Analyst for a retail company. Management wants an interactive dashboard in Excel using Pivot Tables, Pivot Charts, Slicers, Timeline Filters, and Calculated Fields. The dashboard should help analyze sales performance, customer behavior, product performance, country-wise trends, and revenue patterns over time.

The dataset contains the following columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

Calculated Fields & Helper Columns

Students must first create important calculated fields and helper columns:

Revenue: =Quantity*UnitPrice

Month-Year (for trend analysis): =TEXT(InvoiceDate,"MMM-YYYY")

Weekday (to analyze buying patterns): =TEXT(InvoiceDate,"dddd")

Required Dashboard Visuals

Students should build a professional dashboard using Pivot Charts and slicers:

- Monthly Revenue Trend (Line Chart)
- Top 10 Products by Revenue (Bar Chart)
- Country-wise Revenue (Column Chart)
- Top Customers Analysis (Bar Chart)
- Country Contribution (Donut Chart)
- Sales by Weekday (Column/Bar Chart)
- (Optional) Revenue by Hour (if time data exists)

Slicers & Filters

Dashboard must include:

- Country
 - Product (Description)
 - Month-Year
 - Quantity Bucket
 - Timeline Filter (InvoiceDate)
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Insight & Storytelling Guidance (IMPORTANT)

Students are expected not only to create visuals but also to derive meaningful business insights. Each visual should answer a business question.

1. Product-Centric Insights

Focus on product performance:

- Which products generate the highest revenue vs highest quantity?
- Are top-selling products also high-margin contributors?
- Which products are frequently bought together (if possible analysis)?
- Are there “low quantity but high revenue” premium products?

Think: *“What products actually drive business growth?”*

2. Transaction-Centric Insights

Focus on invoice behavior:

- Do we have repeat transactions or bulk purchasing patterns?
- What is the average revenue per invoice?
- Are there spikes in transaction volume on specific days/months?
- Do weekends vs weekdays impact transaction size?

Think: *“How do customers actually buy?”*

3. Customer-Centric Insights

Focus on CustomerID behavior:

- Who are the top revenue-generating customers?
- Do a small group of customers contribute most revenue (Pareto 80/20)?
- Are customers one-time buyers or repeat buyers?
- Which customers purchase high-value baskets?

Think: *“Who is driving the revenue engine?”*

4. Country / Region-Centric Insights

Focus on geography:

- Which countries contribute the highest revenue?
- Is revenue concentrated in a few regions or distributed evenly?
- Which countries have high quantity but low revenue (low-value markets)?
- Which regions show growth potential?

Think: *“Where is the business strongest or weakest?”*

5. Time-Based Insights (Trend Analysis)

Focus on time patterns:

- Is monthly revenue growing, declining, or seasonal?
- Which months show peak performance?
- Which weekdays show highest sales activity?
- Are there sudden spikes or drops in performance?

Think: *“When does the business perform best?”*

Final Dashboard Requirements

The dashboard must:

- Use consistent theme and formatting
 - Avoid clutter and unnecessary colors
 - Include proper alignment and labeling
 - Be interactive with slicers and timeline
 - Be business-focused, not just chart-focused
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Submission Requirements

Students must submit:

- Raw Data Sheet
 - Pivot Tables Sheet
 - Dashboard Sheet
 - 1-Page Business Insights Summary
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Evaluation Criteria

- Correct use of Pivot Tables
- Quality of visual selection
- Dashboard design & aesthetics
- Slicer and filter interactivity
- Calculated fields accuracy
- Depth of business insights (VERY IMPORTANT)